F1 Pitwall: race strategy, measured

Four strategy questions answered from three seasons of OpenF1 timing data, with how far to trust each answer.

Overview | Tyres | Pit stops | Undercut | Model | Data quality | Limits

1. Tyre wear (MEDIUM, s/lap)

+0.045 s/lap

Weak. The fuel correction is assumed (0.04 s/lap), so the slope is mostly that assumption.

2. Green-flag pit loss

23.1 s

Solid under green flags. Safety Car and VSC estimates are unstable.

3. Undercut success rate

44.8%

One definition, small sample; swings with the starting gap. Not a universal rate.

4. LightGBM minus linear (MAE)

-0.150 s

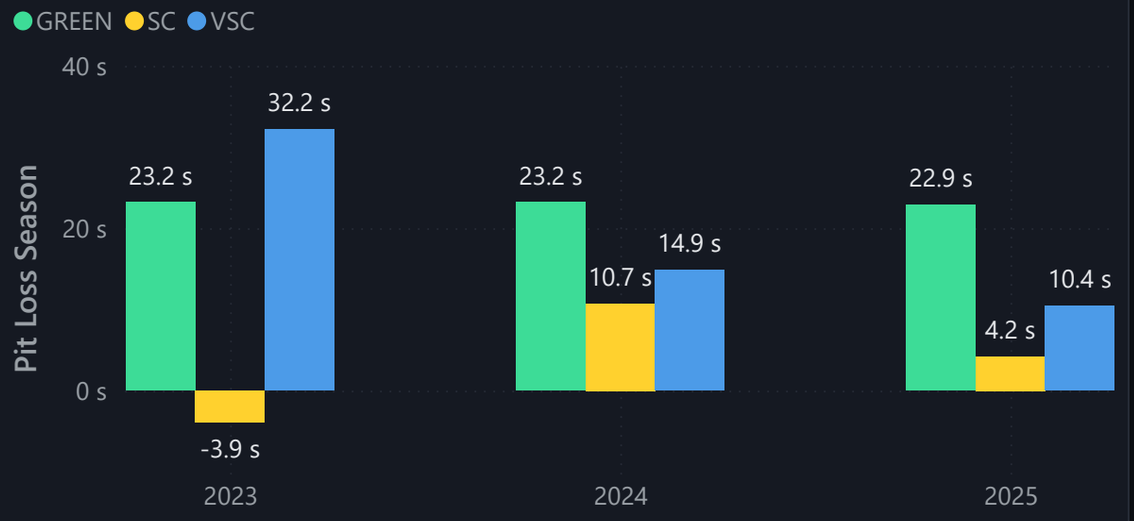
Fixed settings, no tuning; checked across races and whole seasons.

What to take from this

Every figure is computed from validated data and reproducible from the repository. The page-by-page caveats matter as much as the numbers.

- Tyres: wear slopes are descriptive; the assumed fuel effect dominates.
- Pit stops: green-flag loss is steady; Safety Car and VSC loss changes sign between seasons.
- Undercut: the rate depends mostly on the starting gap; car pace is not controlled.
- Model: LightGBM beats the linear baseline on three seasons, not on one.
- Data quality: 23 coded rules, quarantine, and reconciled counts end to end.

Pit loss by season: only green flag is steady



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Tyre degradation

Median within-stint slope per compound, fuel-corrected with an assumed 0.04 s/lap. Filter by season above.

Overview | Tyres | Pit stops | Undercut | Model | Data quality | Limits

season

HARD slope

+0.028 s/lap

HARD, median of 65 race fits

MEDIUM slope

+0.045 s/lap

MEDIUM, median of 79 race fits

SOFT slope

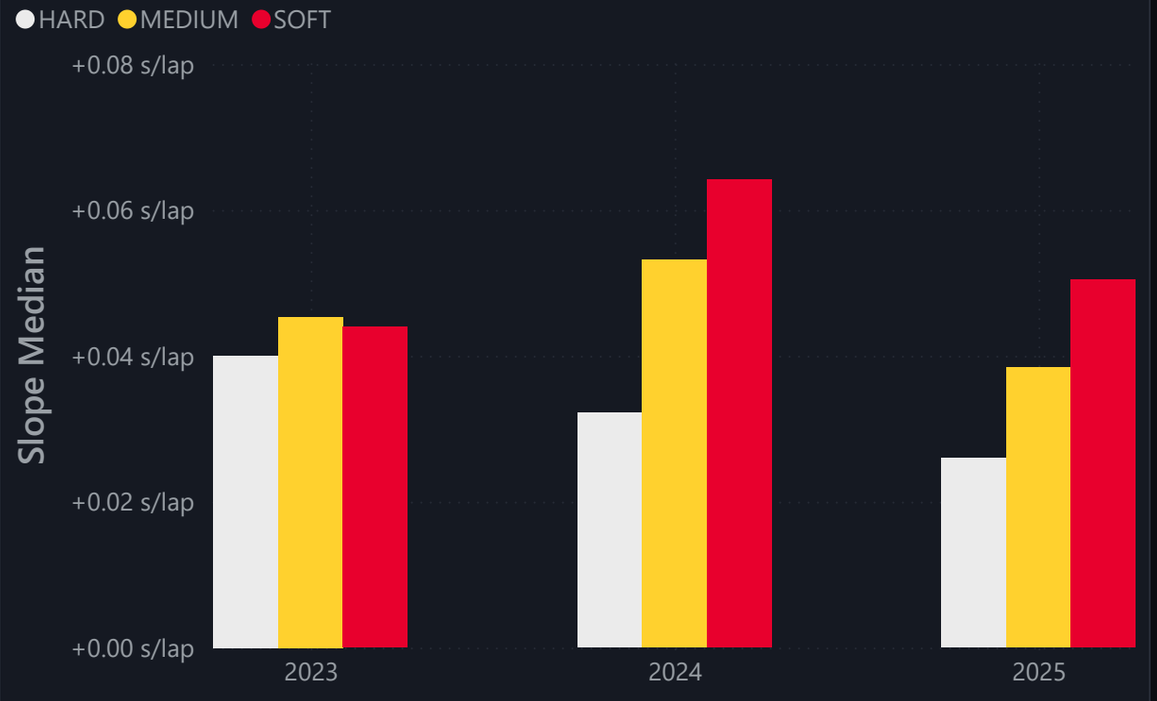
+0.053 s/lap

SOFT, median of 50 race fits

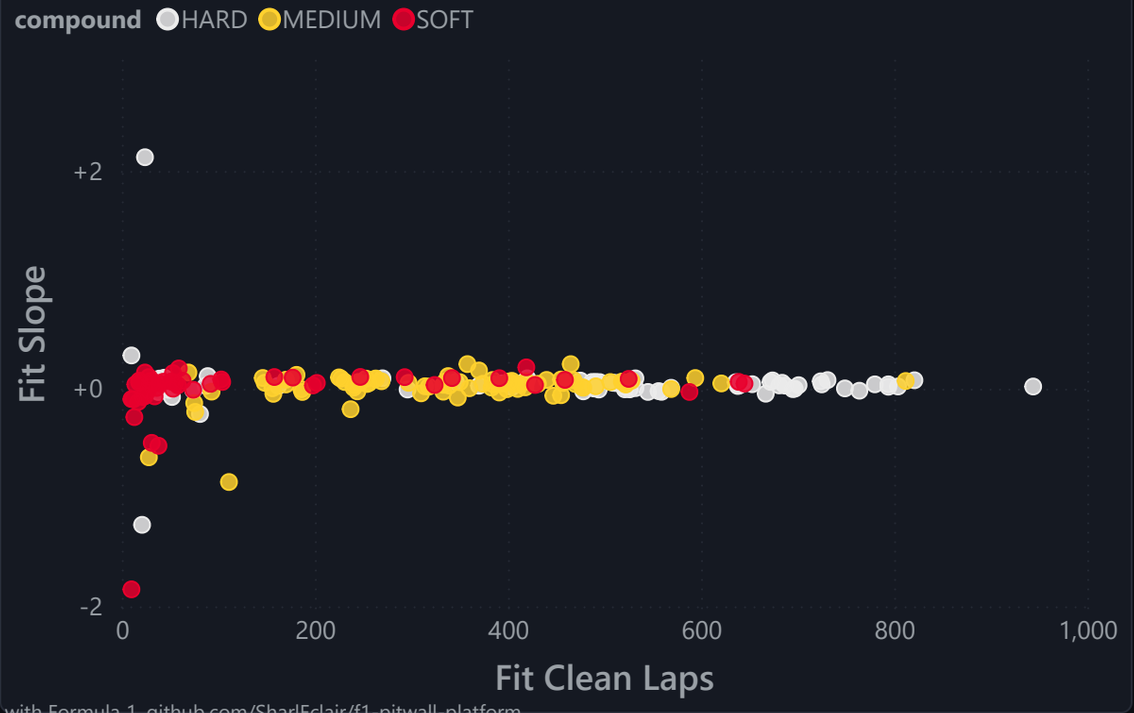
Read with care

The corrected slope is the uncorrected one plus the 0.04 s/lap assumption. Use these to compare compounds and seasons, not as precise rates.

Median slope by season and compound



Each dot is one race fit: small samples scatter most



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Pit-stop cost by track status

In-lap plus out-lap minus two reference laps. Stops over 60 s (red-flag waits) are excluded.

Green flag

23.1 s

Stable across seasons: 22.9 to 23.2 s

Safety Car

2.9 s

Unstable by season: -3.9 to 10.7 s

Virtual Safety Car

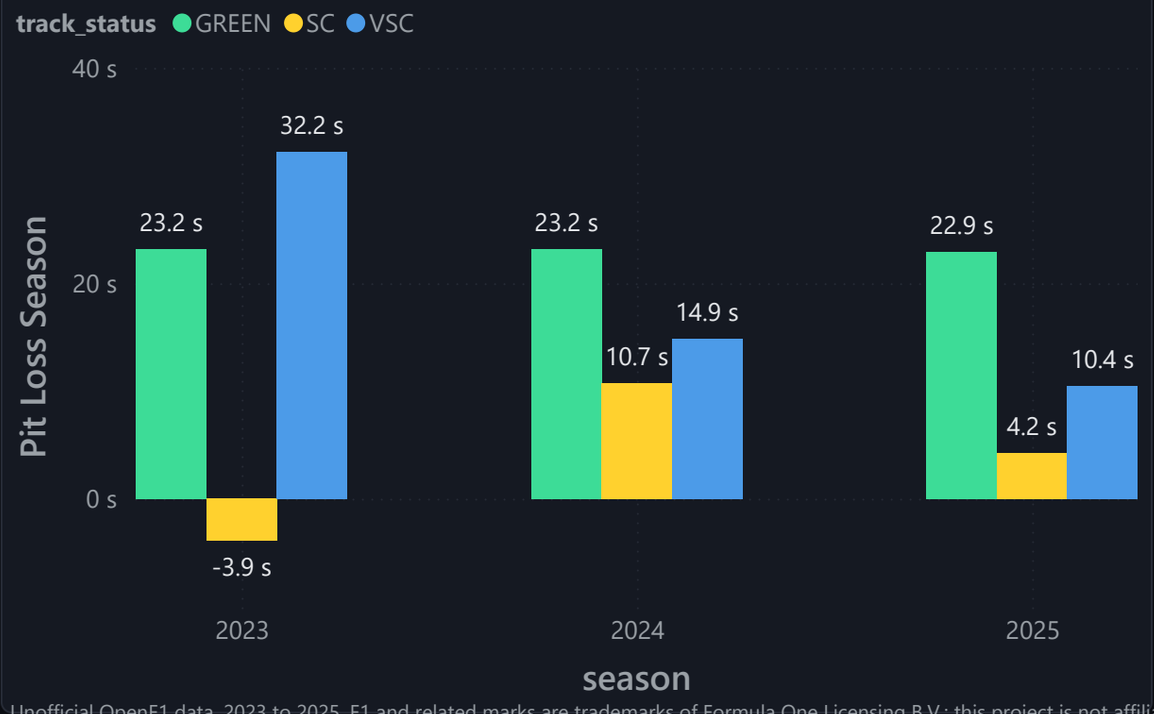
20.9 s

Unstable by season: 10.5 to 32.2 s

Why two of these are not numbers to quote

Under neutralisation the estimator breaks down: the Safety Car value changes sign between seasons. Only the green-flag figure is stable.

Median pit loss by season and status (s)



Season, status, stops and races behind each figure

season	track_status	median_pit_loss_s	stops	races
2023	GREEN	23.17	507	18
2023	SC	-3.89	59	8
2023	VSC	32.17	22	4
2024	GREEN	23.20	619	27
2024	SC	10.71	108	6
2024	VSC	14.87	24	6
2025	GREEN	22.89	490	23
2025	SC	4.21	217	13
2025	VSC	10.45	33	7

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How often does an undercut work?

A frequency under one stated definition, not a universal rate. The definition was fixed before any outcome was computed.

Primary definition: success rate

44.8%

95% interval over races: 35% to 54%

Decided attempts

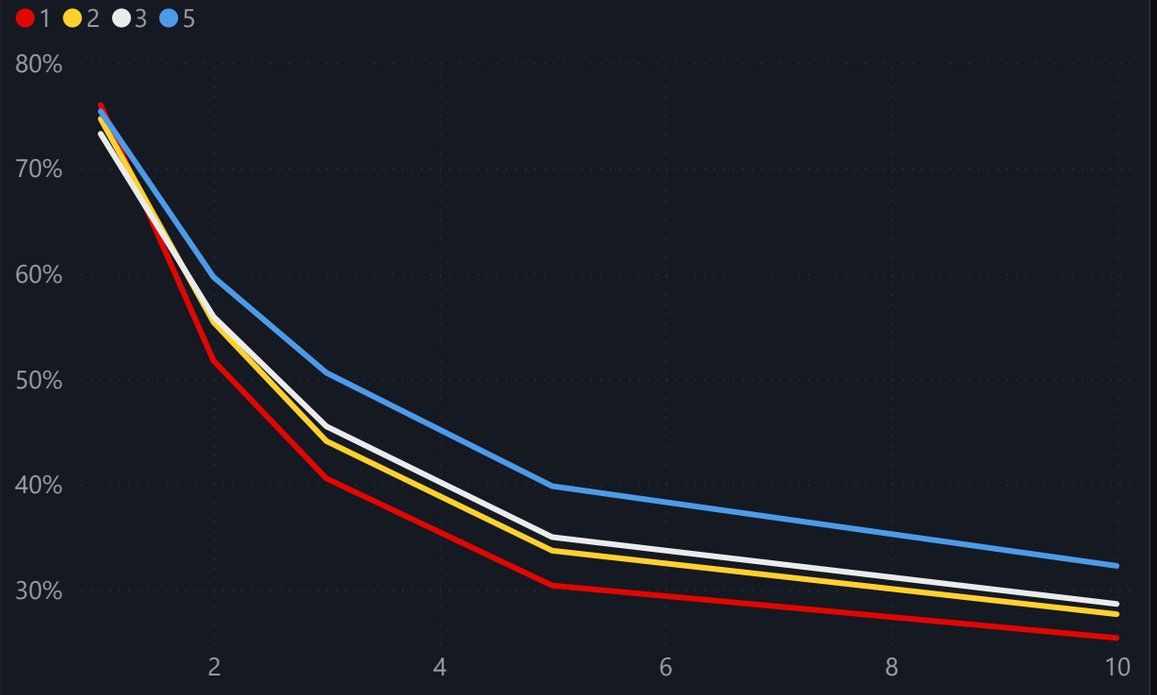
163

164 attempts in 49 races

Primary definition

Car A stops with car B directly ahead and a known gap of at most 3 s; B stops within 3 laps; no Safety Car, VSC or red flag. Success: A is ahead of B two laps after B's stop.

Success rate against the largest starting gap allowed (s)



Sensitivity: success rate by gap (rows) and window (laps)

gap_threshold_s	1	2	3	5	**Total**
1.00	76%	75%	73%	75%	**75%**
2.00	52%	55%	56%	60%	**56%**
3.00	41%	44%	46%	51%	**46%**
5.00	30%	34%	35%	40%	**35%**
10.00	25%	28%	29%	32%	**29%**
Total	**35%**	**38%**	**40%**	**44%**	**40%**

Primary definition by season

season_scope	n_attempts	n_success	n_failure	n_inconclusive	success_rate
2023	48	23	25	0	0.48
2024	66	30	36	0	0.45
2025	50	20	29	1	0.41
all	164	73	90	1	0.45

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Lap-time model: LightGBM against the baselines

Held-out races; every method sees only a driver's first five clean laps of the race. Fixed settings, no tuning.

Overview | Tyres | Pit stops | Undercut | Model | Data quality | Limits

LightGBM minus linear baseline (MAE)

-0.150 s

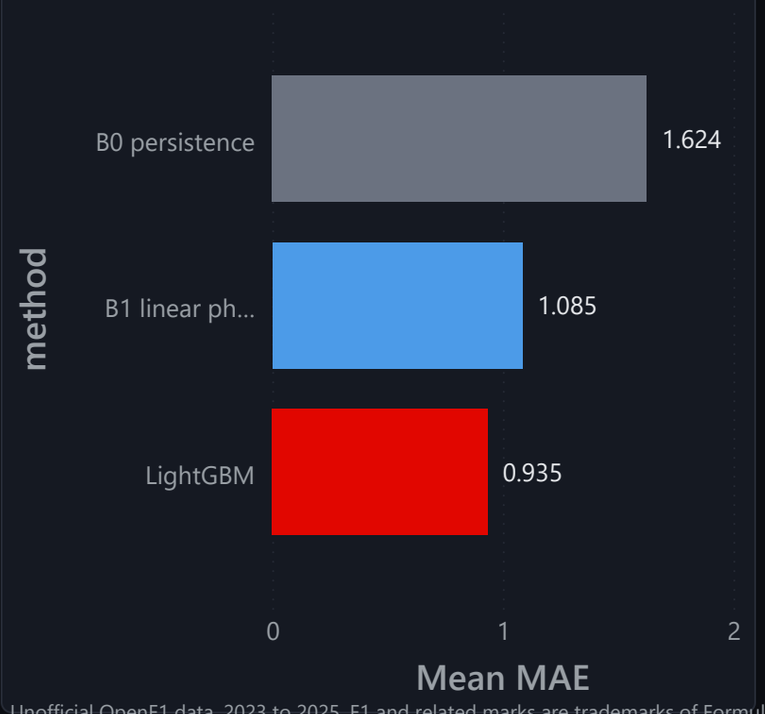
95% interval -0.265 to -0.047 s

Verdict

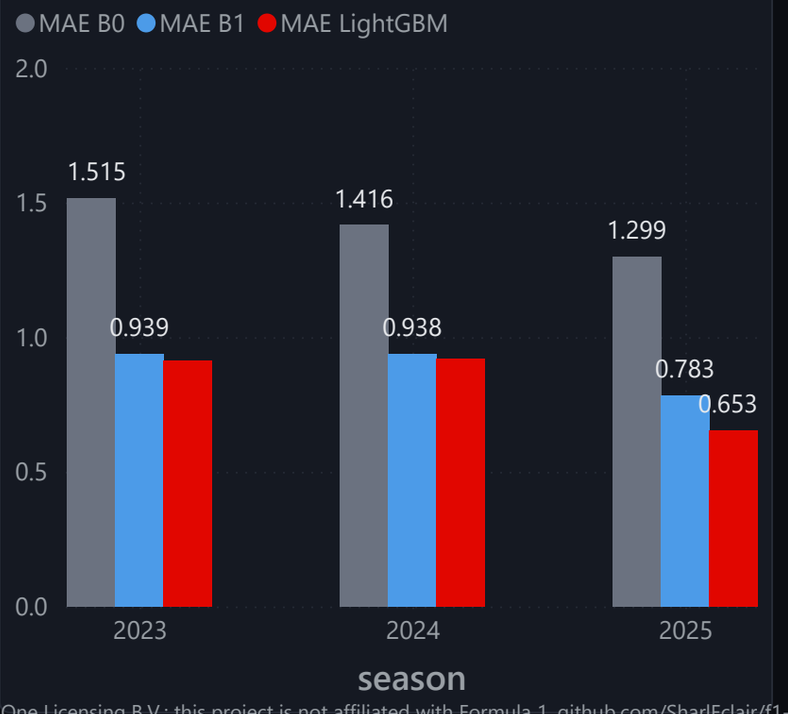
LightGBM better in 47 of 66 races

LightGBM beats the linear baseline: the 95% interval excludes zero.

Mean absolute error across races (s)



Stricter check: hold out a whole season (s)



By race, worst first (LightGBM)

circuit_short_name	season	mae_b1	mae_gbm
Silverstone	2025	7.98	5.31
Monte Carlo	2024	3.01	3.78
Melbourne	2025	4.15	2.80
Monte Carlo	2023	1.82	2.32
Montreal	2024	3.84	2.29
Lusail	2023	1.78	1.82
Zandvoort	2023	2.24	1.81
Mexico City	2024	0.75	1.34
Monza	2023	0.93	1.21
Interlagos	2025	1.15	1.17
Imola	2024	1.18	1.15
Melbourne	2023	1.54	1.06
Spa-Francorchamps	2023	1.19	1.02
Singapore	2023	0.97	0.96
Interlagos	2023	0.85	0.95
Singapore	2025	0.82	0.95
Silverstone	2024	0.82	0.93

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Data quality: what the rules caught

Bad rows are never dropped silently: quarantined rows are kept apart with a reason, flagged rows stay with a marker.

Overview | Tyres | Pit stops | Undercut | Model | Data quality | Limits

Quarantined rows

57

Flagged rows

2,229

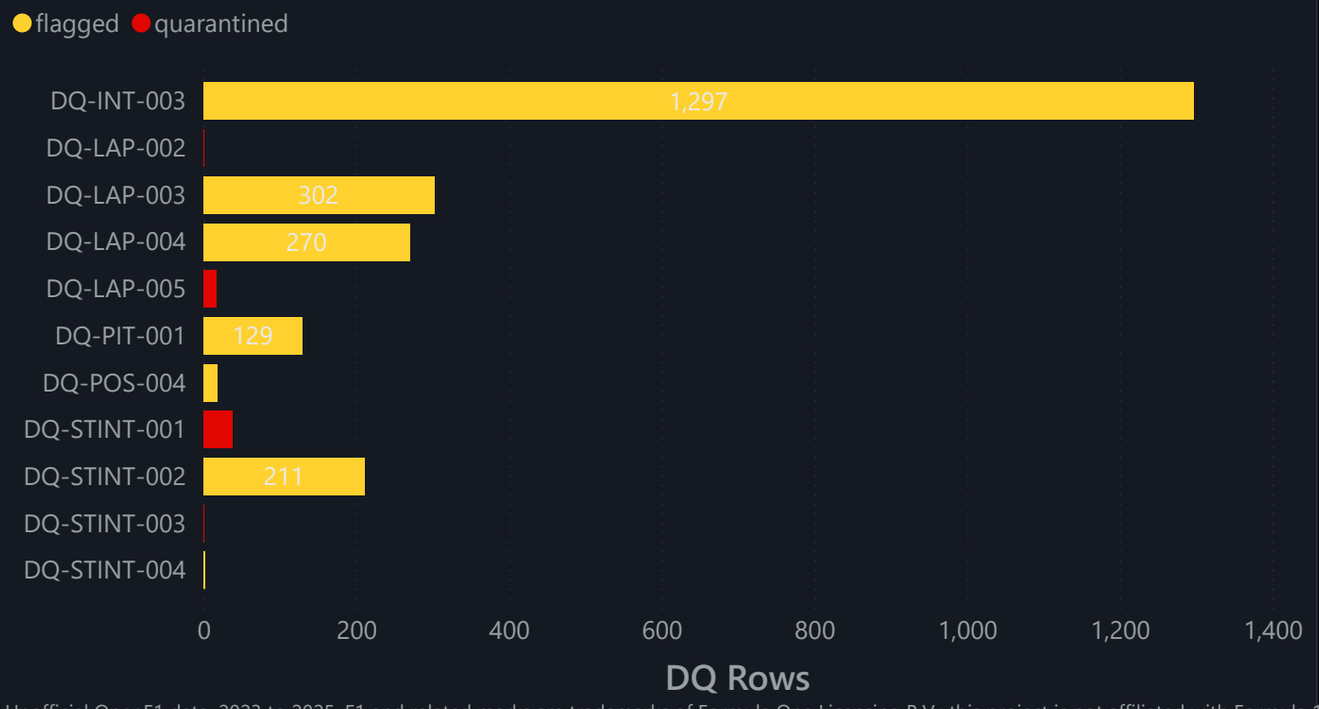
Rules available

23

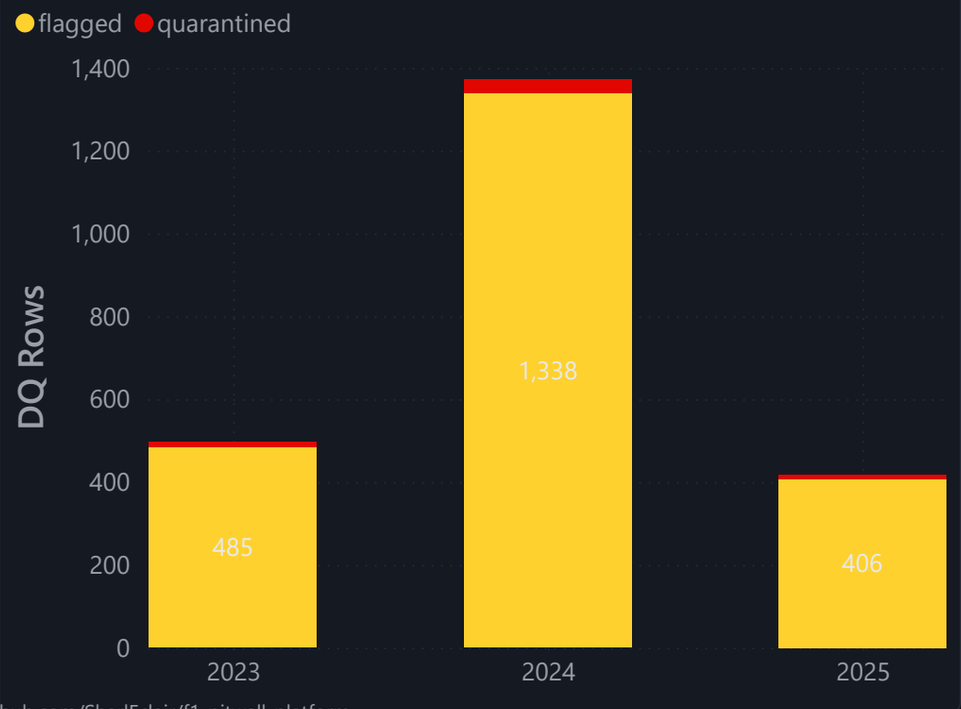
Reconciliation

For every session raw rows = silver rows + quarantined rows, checked in S3 and again after loading into Snowflake.

Rows by rule and outcome (hover for the rule)



Rows by season and outcome



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Method and limits

What each number rests on and what it does not tell you.

Assumptions

- The fuel effect is assumed at 0.04 s per lap.
- Track status comes from message timestamps, so edge laps are approximate.

Estimates

- Pit loss is an estimate from lap times, not stationary time.
- Safety Car and VSC pit loss is unreliable and changes between seasons.

Sample and scope

- Three seasons; nothing here is a claim about other years or regulations.
- Circuits repeat across seasons, so a whole-season check is shown as well.

Not controlled

- Undercut: pace difference between cars, tyres, traffic, DRS and circuit.
- Position is the on-track order, not the official classification.

Data and licence

- Unofficial OpenF1 data; its reuse terms are not stated.
- Only aggregated tables are shared; F1 marks belong to Formula One Licensing B.V.

Blocked or unexplained

- GitHub OIDC deploys are blocked by an organisation policy.
- The Snowflake storage integration never worked; data loads through an internal stage.

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